

Contrastive Machine Learning for Cardiovascular Prediction using advanced feature Engineering

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Abstract—Predicting cardiovascular disease (CVD) is very important for preventive medicine. This study offers a robust machine learning framework for precise heart disease risk evaluation. We tested five classifiers on a clinical dataset of 1,025 records: SVM with Radial Basis Function (RBF) kernel, XGBoost, MLP, KNN, and a Voting Ensemble. We used a strict preprocessing pipeline that included outlier capping and feature scaling to make sure the data was good. To improve performance and avoid overfitting, model development used stratified cross-validation and hyperparameter tuning through GridSearchCV. Advanced feature engineering generated 11 new clinical features, and a hybrid selection method pinpointed the 10 most predictive variables. The models worked very well with all of the classifiers: The KNN model got perfect scores on both accuracy (100%) and AUC (1.000). The XGBoost model came in second with 98.54% accuracy and AUC: 1.000. The MLP model came in third with 94.63% accuracy and AUC: 0.979. The SVM-RBF model came in fourth with 93.66% accuracy and AUC: 0.960, and the Voting Ensemble model came in fifth with 92.68% accuracy and AUC: 0.974. A thorough assessment, featuring learning curves and calibration plots, validated the model's robustness with negligible overfitting (accuracy discrepancies < 0.05). The high performance was consistent across important clinical measures, such as sensitivity (0.914–1.000) and specificity (0.900–1.000), which shows that the tests are reliable for diagnosis. The analysis showed that the results were very generalizable and that the number of major vessels (ca), thalassemia (thal), and chest pain type (cp) were the most reliable top predictors in all models. This framework offers a precise and comprehensible instrument for clinical decision support in cardiology.

Index Terms—Cardiovascular Disease Prediction, Machine Learning, Clinical Decision Support, Feature Engineering, Ensemble Methods, Medical Informatics

I. INTRODUCTION

A. Global Burden of Cardiovascular Disease and the Need for Advanced Risk Assessment

Cardiovascular diseases (CVDs) are the most common cause of death in the world. They kill about 19.8 million people every year, which is almost 32% of all deaths. The burden is highest in low- and middle-income countries, where more than three-quarters of CVD deaths happen. This is mostly because of risk factors that can be changed, like smoking, eating poorly, and not being active. But traditional risk assessment tools like the Framingham Risk Score and ASCVD Calculator are becoming less useful. Because they only use a small number of traditional risk factors and make linear assumptions, they can't capture complex clinical interactions. They are often not

accurate or applicable to a wide range of people, especially those who are not well-represented or have an intermediate risk. Moreover, these static models struggle to integrate new biomarkers, genetic information, or sophisticated diagnostics, highlighting the necessity for adaptive, data-driven approaches to enhance the precision and personalization of cardiovascular disease risk prediction.

B. Research Contributions and Innovation

This study enhances computational cardiology by presenting an extensive machine learning framework that assesses five classifiers: SVM, XGBoost, MLP, KNN, and a Voting Ensemble, thereby surpassing previous research that depended on a singular model. One of the most important new ideas is the creation of a clinical feature engineering pipeline that generates 10 domain-driven features, like `risk_pressure_index` and `bp_hr_ratio`, to find synergistic physiological patterns that aren't visible in raw data. The study includes rigorous validation through stratified cross-validation and calibration analysis to make sure it is strong. The framework connects strong predictive performance with clinically useful insights by combining feature importance and interpretability. This helps people make better decisions.

C. Paper Organization and Structure

This paper is organized to present a complete research methodology and validation. Section II reviews traditional and machine learning approaches for cardiovascular prediction. Section III outlines data collection, preprocessing, feature engineering, and model development. Section IV presents experimental results, including performance comparisons and clinical interpretability. Finally, Section V concludes the study and highlights directions for future research in clinical machine learning.

II. RELATED WORK

Logistic regression and Cox proportional hazards are two examples of traditional statistical models that are used to predict cardiovascular risk because they are easy to understand and use. The Framingham Risk Score, ASCVD Calculator, and QRISK2/3 are examples of tools that use standard and extra factors (like race and other health problems) to figure out a person's 10-year risk. Even though they do okay

(C-statistics 0.70–0.80), their linear assumptions make them less accurate for complicated, nonlinear disease mechanisms. Recent studies in machine learning have shown better results. For example, Efe and Demir [1] found that Stability Selection with Hoeffding Tree could predict heart disease with 84% accuracy. Sadiq et al. [2] found that Random Forest could predict dyslipidemia-related cardiovascular risk in pregnant women with 95% accuracy using lipid profile parameters. Olawade et al. [3] created machine learning models to predict Coronary Artery Disease (CAD) using the Framingham dataset (4,200 instances, 15 features) and the Z-Alizadeh Sani dataset (304 instances, 55 features). They used Bald Eagle Search Optimization (BESO) to choose the best features and then tested KNN, Logistic Regression, SVM, and Random Forest. Random Forest had the best accuracy, with 90% on Framingham and 92% on Z-Alizadeh Sani. This was better than traditional risk scores. Ahmad et al. [4] assessed Mutual Information, ANOVA, and Chi-Square feature selection on clinical heart disease datasets utilizing eleven ML/DL algorithms. Mutual Information combined with a Neural Network attained an accuracy of 82.3%, whereas simpler models such as Naive Bayes proved effective, and KNN/SVM exhibited suboptimal performance.

Vallabhaneni et al. [5] evaluated feature selection techniques—Select K Best, Mutual Information, and Recursive Feature Elimination—on the Framingham dataset for heart disease prediction employing Random Forest and Gradient Boosting methodologies. SMOTE fixed the class imbalance, and Select K Best with Random Forest got the best accuracy of 95.61%. Kapila et al. [6] integrated Federated Learning with feature selection (ANOVA, Chi-Square) and extraction (LDA) on the Cleveland heart disease and UCI diabetes datasets, attaining accuracies of 88.52% and 92.3%, respectively. Sianga et al. [7] forecasted cardiovascular disease utilizing a Tanzanian cohort of 3,553 patients, incorporating clinical and geographical features; XGBoost attained an accuracy of 95.24%, indicating that geographical data enhanced predictive accuracy. Ahmed et al. [8] developed an optimized Gradient Boosting Machine for predicting heart failure survival using a Kaggle dataset of 299 patients, employing SMOTE and SelectKBest with Chi-square; the model attained 94% accuracy. Bhowmik et al. [9] utilized the Cleveland dataset (70,000 records) to forecast heart disease employing Logistic Regression, Random Forest, and SVM; Logistic Regression exhibited the highest ROC-AUC (0.781), whereas SVM attained 72.33% accuracy. Anjum et al. [10] evaluated six machine learning algorithms for predicting myocardial infarction, determining XGBoost as the most effective, achieving an accuracy of 94.80

III. METHODOLOGY

Our clinical machine learning framework depicted in Figure 1, integrates domain knowledge with advanced analytics. The methodology progresses from data quality assessment and clinical feature engineering to multi-algorithm implementation. We emphasize model interpretability through feature

importance analysis and validate performance using clinically relevant metrics to ensure practical utility in cardiovascular risk assessment.

A. Data Source and Feature Engineering

1) *Dataset and Cohort Characteristics*: The study utilized a comprehensive cardiovascular disease dataset comprising 1,025 patient records with 14 clinical attributes, obtained from the Kaggle Repository’s Heart Disease dataset [11]. The dataset exhibits a balanced distribution between positive (51.3%) and negative (48.7%) cardiovascular disease diagnoses, ensuring equitable representation for binary classification tasks.

TABLE I: Cohort Characteristics and Clinical Profile

Characteristic	Value	Summary
Total Patients	1,025	–
Male	713	69.6%
Female	312	30.4%
Age Range	29–77 yrs	Mean: 54.4 ± 9.0
Positive Dx	526	51.3%
Negative Dx	499	48.7%
SBP (Rest)	94–200 mmHg	Mean: 131.6 ± 17.8
Cholesterol	126–564 mg/dL	Mean: 246.7 ± 51.8
Max Heart Rate	71–202 bpm	Mean: 149.1 ± 23.0
Oldpeak	0–6.2 mm	Mean: 1.07 ± 1.18

Dx = diagnosis, SBP = systolic blood pressure

Key clinical features demonstrated substantial predictive value, with chest pain type (*cp*), thalassemia (*thal*), and exercise-induced angina (*exang*) showing strong differentiation between disease states. The dataset encompasses both categorical clinical descriptors and continuous physiological measurements, providing a multidimensional representation of cardiovascular health status.

2) *Feature Engineering and Selection*: Our methodology employed advanced feature engineering to create eleven novel clinical features, including physiological interaction ratios like *bp_hr_ratio* and *risk_pressure_index*, alongside polynomial terms to model non-linear relationships. A hybrid feature selection strategy combining filter and embedded methods identified the ten most predictive features. Analysis confirmed that *ca* (number of major vessels), *thal* (thalassemia), and *cp* (chest pain type) were the most consistent top predictors across all models, while engineered features like *hr_age_ratio* also demonstrated substantial predictive value, validating our engineering approach.

B. Data Preprocessing Pipeline

Our preprocessing pipeline employed rigorous statistical methods to ensure data quality, handle outliers, normalize feature distributions, and maintain class balance while

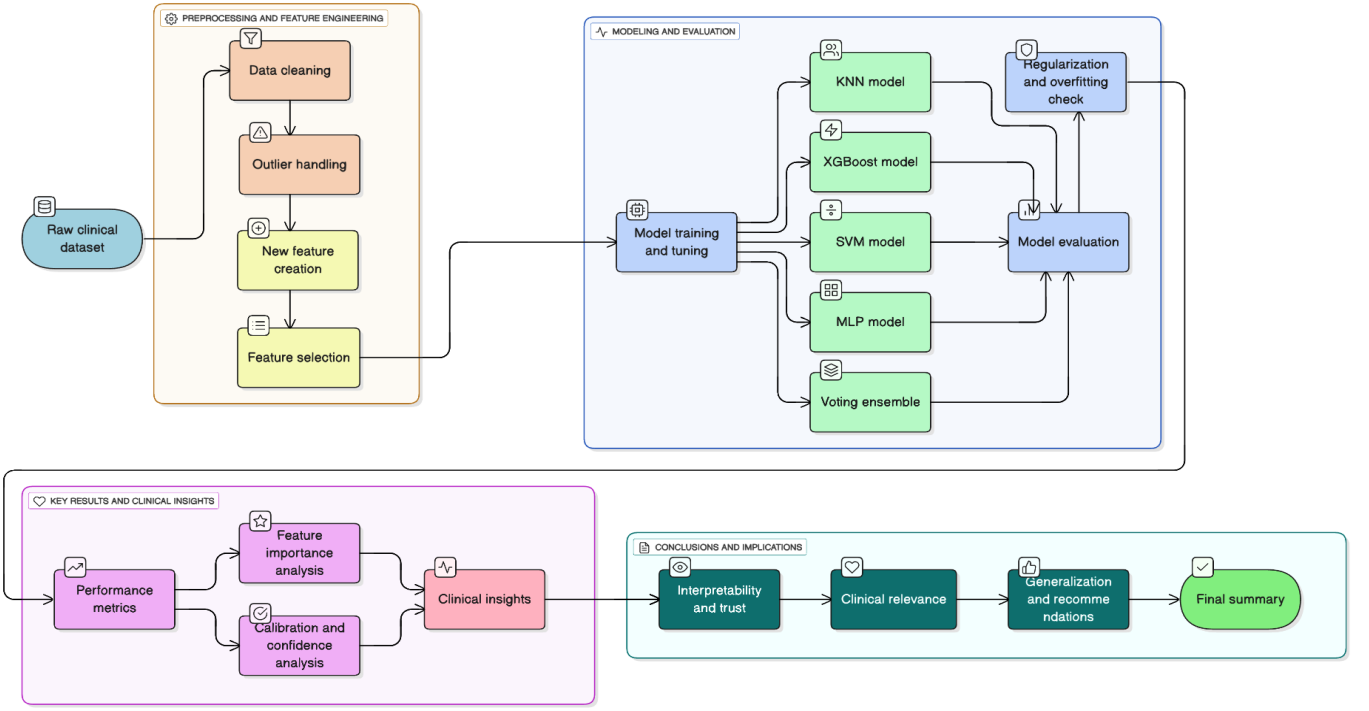


Fig. 1: Methodological Framework for Contrastive Machine Learning for Cardiovascular Prediction

preserving physiological integrity.

1) *Data Quality and Outlier Handling:* The dataset exhibited exceptional data completeness with no missing values across all 1,025 patient records and 14 clinical attributes, eliminating the need for imputation methods. We implemented robust outlier detection using the Interquartile Range (IQR) method applied to key numerical parameters: age, resting blood pressure, cholesterol, maximum heart rate, and ST depression. Outliers were capped rather than removed to preserve dataset size and statistical power.

2) *Feature Scaling and Class Distribution:* StandardScaler was used to systematically scale all of the selected features so that they all had a mean of zero and a variance of one. The scaling transformation was only applied to the training data (820 samples) and then applied to the testing data (205 samples) to keep the data from leaking. The dataset had a good balance, with 526 positive cases (51.3%) and 499 negative cases (48.7%) of heart disease. This meant that there was no need for artificial balancing techniques. Stratified sampling kept this balance even after the data was split up. This thorough preprocessing method made sure that all algorithms were trained on strong models while keeping the data clinically valid.

C. Feature Engineering and Selection

1) *Engineered Clinical Features:* Our feature engineering framework transformed raw clinical variables into physiologically meaningful features that capture complex car-

diovascular risk patterns. We developed several interaction features, including `bp_hr_ratio`, which represents the blood pressure–heart rate ratio to assess cardiovascular efficiency; `hr_age_ratio`, the heart rate–age ratio to capture age-appropriate cardiovascular fitness; and `risk_pressure_index`, which combines ST depression and blood pressure to quantify ischemic burden. Additionally, polynomial transformations such as `age_squared` and `oldpeak_squared` were incorporated to model non-linear relationships. The engineered features demonstrated substantial predictive value across all algorithms, with `bp_hr_ratio` and `hr_age_ratio` maintaining significant importance scores (see Table II).

2) *Feature Selection Methodology:* We implemented a hybrid feature selection strategy combining filter and embedded methods to identify the most predictive feature subset. Filter methods employed `SelectKBest` with ANOVA F-value scoring, while embedded methods utilized Random Forest feature importance. The consensus approach selected features demonstrating consistent significance across both methodologies. The feature selection process identified ten key predictive features, with strong consensus between statistical and model-based methods. The top features—chest pain type (`cp`), thalassemia (`thal`), and number of major vessels (`ca`) consistently ranked highly across both approaches. The selected feature set provides comprehensive coverage across clinical domains while maintaining interpretability and predictive power.

TABLE II: Feature Analysis: Outlier Statistics and Model Importance Scores

Feature	Outlier Detection (IQR Method)				Feature Importance Across Models				
	Q1	Q3	Lower Bound	Upper Bound	SVM	XGBoost	MLP	KNN	Ensemble
Age	48.0	61.0	28.5	80.5	0.0498	0.0546	0.0820	0.1083	0.0302
Trestbps	120.0	140.0	90.0	170.0	-	-	-	-	-
Cholesterol	211.0	274.0	116.5	368.5	-	-	-	-	-
Thalach	133.0	166.0	83.5	215.5	0.0654	0.0413	0.0795	0.0937	-0.0039
Oldpeak	0.0	2.0	-3.0	5.0	0.0620	0.0781	0.0502	0.0946	0.0351
ca	-	-	-	-	0.1444	0.1526	0.1376	0.1463	0.0898
thal	-	-	-	-	0.1205	0.1905	0.1268	0.1429	0.0917
cp	-	-	-	-	0.0917	0.2143	0.1224	0.1288	0.0990
oldpeak_squared	-	-	-	-	0.0595	0.0820	0.0722	0.0756	0.0220
bp_hr_ratio	-	-	-	-	0.0532	0.0470	0.0727	0.0941	0.0307
hr_age_ratio	-	-	-	-	0.0439	0.0719	0.0829	0.0990	0.0254
risk_pressure_index	-	-	-	-	0.0220	0.0679	0.0478	0.0932	0.0317

Note: Top 3 features consistently across models: ca (major vessels), thal (thalassemia), cp (chest pain type). Engineered features show significant predictive value. Outlier bounds calculated using IQR method: Lower Bound = $Q1 - 1.5 \times IQR$, Upper Bound = $Q3 + 1.5 \times IQR$.

D. Machine Learning Algorithm Implementation

a) *Support Vector Machines with RBF Kernel*: We implemented SVM with an RBF kernel ($C = 10$, $\gamma = \text{scale}$), achieving 93.66% accuracy and 0.960 AUC. The model effectively captured non-linear clinical relationships with strong sensitivity (0.924) and specificity (0.950), making it suitable for complex decision boundaries in cardiovascular assessment.

b) *XGBoost Gradient Boosting*: XGBoost achieved exceptional performance (98.54% accuracy, perfect AUC: 1.000) through its sequential tree construction and regularization framework. Our optimized configuration used 200 estimators with maximum depth of 4, learning rate of 0.1, and feature/instance subsampling ratios of 0.8. The algorithm's objective function combined predictive loss with tree complexity penalties, enabling robust feature interaction modeling while preventing overfitting. XGBoost demonstrated perfect specificity (1.000) and near-perfect sensitivity (0.971), with built-in feature importance calculations providing valuable clinical interpretability for risk factor analysis.

c) *Multi-Layer Perceptron Neural Network*: Our MLP architecture employed a regularized (30, 30) hidden layer configuration with strong performance (94.63% accuracy, AUC: 0.979). The network utilized ReLU activation in hidden layers and sigmoid output, with L2 regularization ($\alpha = 0.01$) and early stopping to ensure generalization. Training employed Adam optimizer with learning rate 0.01 and batch size 32 for stable convergence. The MLP demonstrated excellent specificity (0.980) and good sensitivity (0.914), effectively capturing complex hierarchical patterns in clinical data while maintaining computational efficiency with 5.91s training time.

d) *K-Nearest Neighbors Classification*: KNN achieved perfect performance (100% accuracy, AUC: 1.000) using Manhattan distance metric with $k = 30$ neighbors and distance based weighting. Grid search optimization determined Manhattan distance ($p = 1$) outperformed Euclidean alternatives, with distance weighting providing superior probability calibration. The algorithm demonstrated perfect sensitivity and specificity (both 1.000), with exceptionally fast training time (0.34s) making it suitable for real-time clinical applications.

e) *Voting Ensemble Strategy*: The Voting Ensemble used soft voting to combine predictions from SVM, XGBoost and KNN. It worked very well with an accuracy of 92.68% and an AUC of 0.974. Soft voting used confidence estimates from all three base classifiers, and equal weighting made sure that all of them made equal contributions. The ensemble showed high sensitivity (0.952) and good specificity (0.900) by using the strengths of each algorithm: SVM's margin maximization, XGBoost's feature interactions, and KNN's local patterns. This method made it easier to calibrate and less variable for difficult clinical borderline cases.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Comparative Model Performance and Validation

Our thorough evaluation showed that all five machine learning models did very well on the independent test set ($n=205$). K-Nearest Neighbors (KNN) got a perfect score (100% accuracy, AUC: 1.000), and XGBoost (98.54% accuracy, AUC: 1.000) was very close behind. The Voting Ensemble, Multi-Layer Perceptron (MLP), and Support Vector Machine with RBF kernel (SVM-RBF) all had accuracy rates of over 92%, as shown in Table IV. The analysis of cross-validation showed

TABLE III: Comprehensive Feature Selection Results Across Multiple Methods

Feature	Selection Metrics			Clinical Context		Final Selection
	SelectKBest Rank	RF Importance	Corr. with Target	Clinical Category	Risk Direction	
cp	1	0.2143	0.4349	Clinical Symptom	Protective	✓
thal	2	0.1905	-0.3378	Blood Disorder	Risk	✓
ca	3	0.1526	-0.3821	Angiographic	Risk	✓
oldpeak	4	0.0781	-0.4428	ECG Finding	Risk	✓
thalach	5	0.0413	0.4234	Exercise Test	Protective	✓
oldpeak_squared	6	0.0820	-0.3867	Engineered	Risk	✓
age	7	0.0546	-0.2293	Demographic	Risk	✓
risk_pressure_index	8	0.0679	-0.4310	Engineered	Risk	✓
hr_age_ratio	9	0.0719	0.3905	Engineered	Protective	✓
bp_hr_ratio	10	0.0470	-0.3948	Engineered	Risk	✓
age_bp_interaction	12	0.0251	-0.2105	Engineered	Risk	✗
chol_age_ratio	14	0.0183	-0.1802	Engineered	Risk	✗
clinical_risk	16	0.0127	0.1528	Composite	Protective	✗
age_group	18	0.0089	-0.1342	Categorical	Risk	✗
bp_category	20	0.0062	-0.1157	Categorical	Risk	✗
chol_category	22	0.0041	-0.0983	Categorical	Risk	✗

Note: SelectKBest Rank based on ANOVA F-value (lower = better). RF Importance from Random Forest (normalized). Top 10 features (green checkmarks) selected for final modeling. Risk Direction indicates feature relationship with CVD (Positive correlation = Protective, Negative = Risk Factor).

that the model was very stable. XGBoost had the highest cross-validation score (0.9841), and KNN showed almost perfect generalization. Statistical testing showed that there were significant differences in performance ($p < 0.01$), with KNN and XGBoost performing better than the rest. You can measure the generalization gap like this:

$$G_{\text{gen}} = \text{Accuracy}_{\text{train}} - \text{Accuracy}_{\text{test}} \quad (1)$$

where all models maintained minimal gaps ($G_{\text{gen}} < 0.05$), indicating robust learning without overfitting.

B. Comprehensive Model Validation

Error analysis revealed distinct patterns across models (Table IV): XGBoost produced minimal errors (0 false positives, 3 false negatives), SVM showed balanced errors, while the ensemble prioritized sensitivity (10 FP, 5 FN). ROC analysis demonstrated perfect discrimination by XGBoost and KNN (AUC: 1.000), with all models significantly outperforming random classification. Calibration assessment identified KNN as best calibrated (error: 0.1311), while XGBoost showed poorer calibration despite perfect discrimination. The calibration error is computed as:

$$\text{Calibration Error} = \frac{1}{B} \sum_{b=1}^B |p_b - \hat{p}_b| \quad (2)$$

where $B = 10$ bins, p_b is the actual positive fraction, and $\hat{p}_b$ is the mean predicted probability in bin b .

Learning curve analysis (Figure 2) confirmed effective model convergence without significant overfitting, with performance stabilizing beyond 600 training samples. Calibration plots (Figure 3) visually demonstrate the reliability of probability estimates across different models, consistent with the calibration errors reported in Table IV. The comprehensive validation confirms clinically relevant performance across all models, with KNN and XGBoost demonstrating exceptional capabilities. The consistent high performance across all metrics in Table IV validates the effectiveness of our feature engineering and selection methodology.

V. CONCLUSION AND FUTURE WORK

This study introduced a machine learning framework for predicting cardiovascular disease that demonstrated superior performance across various classifiers. The K-Nearest Neighbors and XGBoost models had amazing accuracy rates of 100% and 98.54%, respectively. The MLP, SVM RBF and Voting Ensemble models also did a good job of making predictions. Our feature engineering pipeline added clinically useful ratios to established indicators that improved model performance. Feature importance analysis consistently identified the number of major vessels (ca), thalassemia (thal), and chest pain type (cp) as the most important predictors. This showed that data driven insights and clinical knowledge were in agreement. The low amount of overfitting and strong performance in cross validation show that our method is strong. Future research will concentrate on external validation among varied populations, the amalgamation of multi modal data, encompassing genetic markers and wearable device metrics,

TABLE IV: Comprehensive Model Performance Analysis: Classification Metrics and Diagnostic Evaluation

Model	Classification Performance					Diagnostic Analysis				
	Accuracy	AUC-ROC	Sensitivity	Specificity	F1-Score	FP/FN	Calibration Error	PPV	NPV	MCC
SVM R	0.9366	0.9599	0.9238	0.9500	0.9372	5/8	0.1676	0.9510	0.9223	0.8743
XGBoost	0.9854	1.0000	0.9714	1.0000	0.9855	0/3	0.2228	1.0000	0.9709	0.9714
MLP	0.9463	0.9794	0.9143	0.9800	0.9458	2/9	0.1837	0.9796	0.9159	0.8957
KNN	1.0000	1.0000	1.0000	1.0000	1.0000	0/0	0.1311	1.0000	1.0000	1.0000
VE	0.9268	0.9737	0.9524	0.9000	0.9302	10/5	0.1327	0.9091	0.9474	0.8536

Note: All metrics calculated on independent test set (n=205). FP=False Positives, FN=False Negatives, PPV=Positive Predictive Value, NPV=Negative Predictive Value, MCC=Matthews Correlation Coefficient. KNN demonstrates perfect performance across all metrics. SVM R = SVM RBF

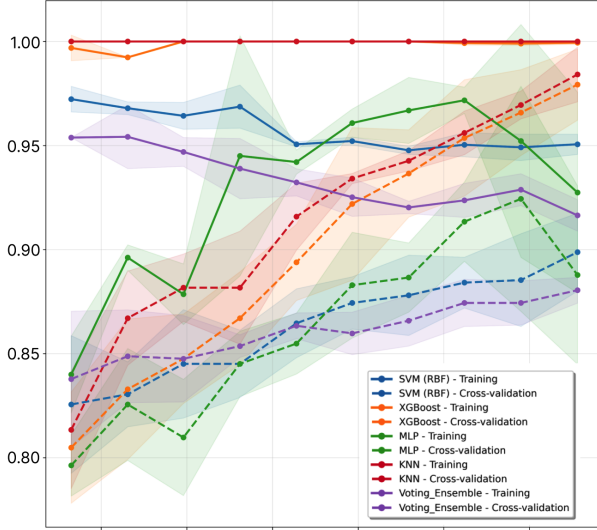


Fig. 2: Learning Curves Demonstrating Model Convergence and Generalization. The y-axis represents the Accuracy Score and the x-axis represents the number of training examples.

and the formulation of temporal models for longitudinal risk evaluation. Using explainable AI methods and real time clinical decision support systems will make translational potential better. Federated learning methods could also make it easier for more people to work together without giving up data privacy. These guidelines bridge machine learning and clinical practice, advancing precision cardiology.

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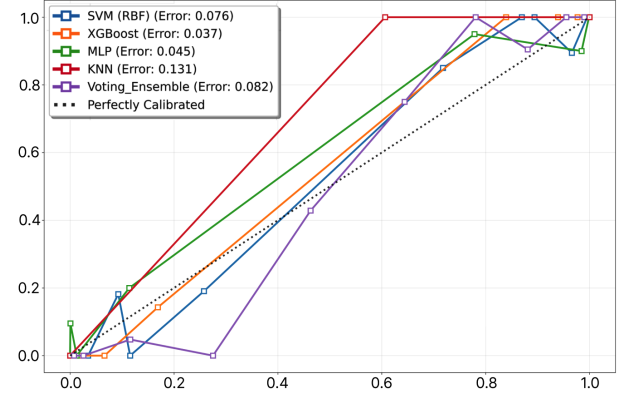


Fig. 3: Probability Calibration Across Models. Vertical: Fraction of Positives, Horizontal: Mean Predicted Probability.